535240175_Pizza_Classification_P7

April 13, 2026

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0.0.2 Notebook Imports

[1]:
```
import os
import pathlib
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.image as mpimg
import seaborn as sns
sns.set()

import warnings
warnings.filterwarnings("ignore")

import tensorflow as tf
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.layers import Dense, Dropout, Flatten, Conv2D, MaxPool2D
from tensorflow.keras import Sequential, Input
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.losses import BinaryCrossentropy
from tensorflow.keras import layers, models
import tensorflow_hub as hub
```

```
WARNING: All log messages before absl::InitializeLog() is called are written to
STDERR
I0000 00:00:1776065546.785047   16511 port.cc:153] oneDNN custom operations are
on. You may see slightly different numerical results due to floating-point
round-off errors from different computation orders. To turn them off, set the
```

```
environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
I0000 00:00:1776065546.832722   16511 cpu_feature_guard.cc:227] This TensorFlow
binary is optimized to use available CPU instructions in performance-critical
operations.
To enable the following instructions: AVX2 AVX512F AVX512_VNNI FMA, in other
operations, rebuild TensorFlow with the appropriate compiler flags.
WARNING: All log messages before absl::InitializeLog() is called are written to
STDERR
I0000 00:00:1776065548.032727   16511 port.cc:153] oneDNN custom operations are
on. You may see slightly different numerical results due to floating-point
round-off errors from different computation orders. To turn them off, set the
environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
```

0.0.3 1. Importing Data

[2]:
```
# install kaggle
!pip install kaggle

#saya pakai vs code, jadi lgsg upload kaggle di folder
!mkdir -p ~/.kaggle
!cp kaggle.json ~/.kaggle/
!chmod 600 ~/.kaggle/kaggle.json
```

```
Requirement already satisfied: kaggle in /home/kaming/Work/data-
science/.venv/lib/python3.12/site-packages (2.0.0)
Requirement already satisfied: bleach in /home/kaming/Work/data-
science/.venv/lib/python3.12/site-packages (from kaggle) (6.3.0)
Requirement already satisfied: kagglesdk<1.0,>=0.1.15 in /home/kaming/Work/data-
science/.venv/lib/python3.12/site-packages (from kaggle) (0.1.16)
Requirement already satisfied: packaging in /home/kaming/Work/data-
science/.venv/lib/python3.12/site-packages (from kaggle) (26.0)
Requirement already satisfied: protobuf in /home/kaming/Work/data-
science/.venv/lib/python3.12/site-packages (from kaggle) (7.34.1)
Requirement already satisfied: python-dateutil in /home/kaming/Work/data-
science/.venv/lib/python3.12/site-packages (from kaggle) (2.9.0.post0)
Requirement already satisfied: python-slugify in /home/kaming/Work/data-
science/.venv/lib/python3.12/site-packages (from kaggle) (8.0.4)
Requirement already satisfied: requests in /home/kaming/Work/data-
science/.venv/lib/python3.12/site-packages (from kaggle) (2.33.1)
Requirement already satisfied: tqdm in /home/kaming/Work/data-
science/.venv/lib/python3.12/site-packages (from kaggle) (4.67.3)
Requirement already satisfied: urllib3>=1.15.1 in /home/kaming/Work/data-
science/.venv/lib/python3.12/site-packages (from kaggle) (2.6.3)
Requirement already satisfied: webencodings in /home/kaming/Work/data-
science/.venv/lib/python3.12/site-packages (from bleach->kaggle) (0.5.1)
Requirement already satisfied: six>=1.5 in /home/kaming/Work/data-
science/.venv/lib/python3.12/site-packages (from python-dateutil->kaggle)
(1.17.0)
```

```
Requirement already satisfied: text-unidecode>=1.3 in /home/kaming/Work/data-
science/.venv/lib/python3.12/site-packages (from python-slugify->kaggle) (1.3)
Requirement already satisfied: charset_normalizer<4,>=2 in
/home/kaming/Work/data-science/.venv/lib/python3.12/site-packages (from
requests->kaggle) (3.4.6)
Requirement already satisfied: idna<4,>=2.5 in /home/kaming/Work/data-
science/.venv/lib/python3.12/site-packages (from requests->kaggle) (3.11)
Requirement already satisfied: certifi>=2023.5.7 in /home/kaming/Work/data-
science/.venv/lib/python3.12/site-packages (from requests->kaggle) (2026.2.25)
```

[3]:
```
!kaggle datasets download -d carlosrunner/pizza-not-pizza
```

```
Warning: Looks like you're using an outdated `kaggle` version (installed:
2.0.0), please consider upgrading to the latest version (2.0.1)
Dataset URL: https://www.kaggle.com/datasets/carlosrunner/pizza-not-pizza
License(s): other
pizza-not-pizza.zip: Skipping, found more recently modified local copy (use
--force to force download)
```

[ ]:
```
!unzip -o pizza-not-pizza.zip
```

[6]:
```
data_directory = pathlib.Path("pizza_not_pizza")
# mengambil nama kelas
class_names = [item.name for item in data_directory.glob('*')]
print(class_names)
# path masing-masing kelas
pizza_dir = data_directory / "pizza"
not_pizza_dir = data_directory / "not_pizza"
```

```
['not_pizza', 'food101_subset.py', 'pizza']
```

Let us first check how many images we have of a pizza

[7]:
```
num_pizza_images = len(os.listdir(pizza_dir))
non_pizza_images = len(os.listdir(not_pizza_dir))
print(f'Number of Pizza images: {num_pizza_images}')
print(f'Number of Non-Pizza images: {non_pizza_images}')
```

```
Number of Pizza images: 983
Number of Non-Pizza images: 983
```

0.0.4 2. Visualizing the images

[8]:
```
data_gen = ImageDataGenerator(rescale = 1/255., validation_split = 0.2)

train_data = data_gen.flow_from_directory(
    data_directory,
    target_size = (224, 224),
    batch_size = 32,
    subset = 'training',
```

```
    class_mode = 'binary')
val_data = data_gen.flow_from_directory(
    data_directory,
    target_size = (224, 224),
    batch_size = 32,
    subset = 'validation',
    class_mode = 'binary')
```

```
Found 1574 images belonging to 2 classes.
Found 392 images belonging to 2 classes.
```

[9]:
```
images, labels = next(train_data)
len(images), len(labels), images[0].shape
```

[9]:
```
(32, 32, (224, 224, 3))
```

[10]:
```
def plot_random_images():
    """
    Plots 20 Random Images from the dataset
    """
    images, labels = next(train_data)
    labels = labels.astype('int32')
    i = 1

    plt.figure(figsize = (10, 5))

    for image, label in zip(images, labels):
        plt.subplot(4, 5, i)
        plt.imshow(image)
        plt.title(class_names[label])
        plt.axis('off')
        i += 1
        if i == 21:
            break
    plt.tight_layout()
    plt.show()
```

[11]:
```
plot_random_images()
```

0.0.5 3. Baseline model

Let Us First create a baseline model without doing data augmentation and not adding a Max Pooling Layer

[12]:
```
tf.random.set_seed(42)

model_1 = Sequential([
    Input(shape = (224, 224, 3)), # Input layer
    Conv2D(filters = 32, kernel_size = 2, padding = 'valid', activation =␣
 ↪'relu'), # First C
    Conv2D(filters = 32, kernel_size = 2, padding = 'valid', activation =␣
 ↪'relu'), # Second
    Flatten(),
    Dense(1, activation = 'sigmoid') # Output layer
])
model_1.compile(loss = BinaryCrossentropy(),
                optimizer = Adam(),
                metrics = ['accuracy'])

history_1 = model_1.fit(train_data,
                        epochs= 5,
                        steps_per_epoch = len(train_data),
                        validation_data = val_data,
                        validation_steps = len(val_data))
```

```
I0000 00:00:1776065574.695712   16511 gpu_device.cc:2043] Created device
/job:localhost/replica:0/task:0/device:GPU:0 with 2610 MB memory:  -> device: 0,
```

```
name: NVIDIA GeForce GTX 1650 with Max-Q Design, pci bus id: 0000:01:00.0,
compute capability: 7.5

Epoch 1/5

I0000 00:00:1776065575.651708   16511 generator_dataset_op.cc:213] Memory patch
applied: M_TRIM_THRESHOLD=128 kb was set.
I0000 00:00:1776065576.379892   16732 service.cc:153] XLA service 0x754324031ba0
initialized for platform CUDA (this does not guarantee that XLA will be used).
Devices:
I0000 00:00:1776065576.379912   16732 service.cc:161]   StreamExecutor [0]:
NVIDIA GeForce GTX 1650 with Max-Q Design, Compute Capability 7.5 (Driver:
13.0.0; Runtime: 12.9.0; Toolkit: 12.5.0; DNN: 9.20.0)
I0000 00:00:1776065576.398889   16732 dump_mlir_util.cc:269] disabling MLIR
crash reproducer, set env var `MLIR_CRASH_REPRODUCER_DIRECTORY` to enable.
I0000 00:00:1776065576.598113   16732 cuda_dnn.cc:461] Loaded cuDNN version
92000

 2/50              3s 72ms/step - accuracy:
0.5156 - loss: 1.1534

I0000 00:00:1776065583.246091   16732 device_compiler.h:208] Compiled cluster
using XLA!  This line is logged at most once for the lifetime of the process.

50/50              19s 235ms/step -
accuracy: 0.5534 - loss: 2.2074 - val_accuracy: 0.6913 - val_loss: 0.6681
Epoch 2/5
50/50              4s 87ms/step -
accuracy: 0.7154 - loss: 0.5919 - val_accuracy: 0.6684 - val_loss: 0.5998
Epoch 3/5
50/50              4s 88ms/step -
accuracy: 0.8463 - loss: 0.3911 - val_accuracy: 0.6505 - val_loss: 0.6212
Epoch 4/5
50/50              4s 87ms/step -
accuracy: 0.9562 - loss: 0.1627 - val_accuracy: 0.6811 - val_loss: 0.6803
Epoch 5/5
50/50              4s 85ms/step -
accuracy: 0.9892 - loss: 0.0574 - val_accuracy: 0.7296 - val_loss: 0.6757
```

0.0.6 4. Visualizing the Results

[13]:
```python
# Plot the validation and training data separately
def plot_loss_curves(history):
    """
    Plots the curves of both loss and accuracy
    """

    loss = history.history['loss']
    val_loss = history.history['val_loss']
```

```
    accuracy = history.history['accuracy']
    val_accuracy = history.history['val_accuracy']

    epochs = range(len(loss))

    fig, ax = plt.subplots(1, 2, figsize = (20, 5))

    # Plotting loss
    ax1 = sns.lineplot(x = epochs, y = loss, label='Training Loss', ax= ax[0])
    ax1 = sns.lineplot(x = epochs, y = val_loss, label='Validation Loss', ax=␣
↪ax[0])
    ax1.set(title = 'Loss', xlabel = 'Epochs')

    # Plot accuracy
    ax2 = sns.lineplot(x = epochs, y = accuracy, label='Training Accuracy', ax=␣
↪ax[1])
    ax2 = sns.lineplot(x = epochs, y = val_accuracy, label='Validation␣
↪Accuracy', ax=ax[1])
    ax2.set(title = 'Accuracy', xlabel = 'Epochs')
```

[14]:
```
plot_loss_curves(history_1)
```

0.0.7 5. Adding a Max Pooling Layer

Lets add a Max Pooling layer and check whether the accuracy improves

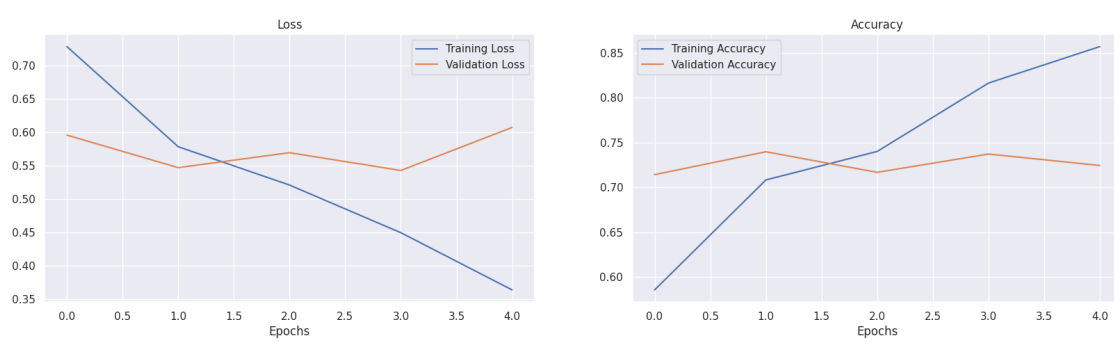
[15]:
```
model_2 = Sequential([
    Input(shape = (224, 224, 3)),
    Conv2D(filters = 32, kernel_size = 2, padding = 'valid', activation =␣
↪'relu'),
    MaxPool2D(pool_size = 2), # It will reduce the number of features by half
    Conv2D(filters = 32, kernel_size =2, padding = 'valid', activation =␣
↪'relu'),
    MaxPool2D(pool_size = 2),
```

```
    Flatten(),
    Dense(1, activation = 'sigmoid')
])
model_2.compile(
    loss = BinaryCrossentropy(),
    optimizer = Adam(),
    metrics = ['accuracy'])

history_2 = model_2.fit(
    train_data,
    epochs= 5,
    steps_per_epoch = len(train_data),
    validation_data = val_data,
    validation_steps = len(val_data))
```

```
Epoch 1/5
50/50              11s 157ms/step -
accuracy: 0.5858 - loss: 0.7281 - val_accuracy: 0.7143 - val_loss: 0.5957
Epoch 2/5
50/50              4s 76ms/step -
accuracy: 0.7084 - loss: 0.5783 - val_accuracy: 0.7398 - val_loss: 0.5469
Epoch 3/5
50/50              4s 87ms/step -
accuracy: 0.7402 - loss: 0.5209 - val_accuracy: 0.7168 - val_loss: 0.5693
Epoch 4/5
50/50              4s 77ms/step -
accuracy: 0.8164 - loss: 0.4496 - val_accuracy: 0.7372 - val_loss: 0.5428
Epoch 5/5
50/50              4s 72ms/step -
accuracy: 0.8571 - loss: 0.3640 - val_accuracy: 0.7245 - val_loss: 0.6071
```

[16]:
```
plot_loss_curves(history_2)
```



We have a maximum validation accuracy of around 75% which is an improvement as compared to our previous model, let us do data augmentation and see whether it improves the accuracy further

0.0.8 6. Data Augmentation

[17]:
```
data_gen_augmented = ImageDataGenerator(
    rescale = 1/255.,
    validation_split = 0.2,
    zoom_range = 0.2,
    horizontal_flip= True,
    rotation_range = 20,
    width_shift_range=0.2,
    height_shift_range=0.2)

print('Augmented training Images:')
train_data_augmented = data_gen_augmented.flow_from_directory(
    data_directory,
    target_size = (224, 224),
    batch_size = 32,
    subset = 'training',
    class_mode = 'binary')

print('Unchanged Validation Images:')
val_data = data_gen.flow_from_directory(
    data_directory,
    target_size = (224, 224),
    batch_size = 32,
    subset = 'validation',
    class_mode = 'binary')
```

```
Augmented training Images:
Found 1574 images belonging to 2 classes.
Unchanged Validation Images:
Found 392 images belonging to 2 classes.
```

[18]:
```
def plot_random_images_augmented():
    """
    Plots 20 Random Images from the dataset
    """

    images_augmented, labels_augmented = next(train_data_augmented)
    labels_augmented = labels_augmented.astype('int32')
    i = 1
    plt.figure(figsize = (10, 5))

    for image, label in zip(images_augmented, labels_augmented):
        plt.subplot(4, 5, i)
        plt.imshow(image)
        plt.title(class_names[label])
        plt.axis('off')
        i += 1
```

```
        if i == 21:
            break

    plt.tight_layout()
    plt.show()
```

[19]:
```
plot_random_images()
print('==========================================================================================
plot_random_images_augmented()
```

```
==========================================================================================
==================
```

[20]:
```
model_3 = Sequential([
    Input(shape = (224, 224, 3)),
    Conv2D(filters = 32, kernel_size = 2, padding = 'valid', activation =␣
 ↪'relu'),
    MaxPool2D(pool_size = 2),
    Conv2D(filters = 32, kernel_size = 2, padding = 'valid', activation =␣
 ↪'relu'),
    MaxPool2D(pool_size = 2),
    Flatten(),
    Dense(1, activation = 'sigmoid')
    ])

model_3.compile(
    loss = BinaryCrossentropy(),
    optimizer = Adam(),
    metrics = ['accuracy'])

history_3 = model_3.fit(
    train_data_augmented, # changing the training data
    epochs= 5,
    steps_per_epoch = len(train_data_augmented),
    validation_data = val_data,
    validation_steps = len(val_data))
```

```
Epoch 1/5
50/50              16s 295ms/step -
accuracy: 0.5521 - loss: 0.7440 - val_accuracy: 0.6658 - val_loss: 0.6516
Epoch 2/5
```

```
50/50              14s 282ms/step -
accuracy: 0.6385 - loss: 0.6344 - val_accuracy: 0.6276 - val_loss: 0.6679
Epoch 3/5
50/50              13s 259ms/step -
accuracy: 0.6347 - loss: 0.6413 - val_accuracy: 0.6939 - val_loss: 0.6027
Epoch 4/5
50/50              14s 274ms/step -
accuracy: 0.6703 - loss: 0.6108 - val_accuracy: 0.7092 - val_loss: 0.5866
Epoch 5/5
50/50              13s 262ms/step -
accuracy: 0.6842 - loss: 0.5959 - val_accuracy: 0.7296 - val_loss: 0.5872
```

[21]:
```
plot_loss_curves(history_3)
```

We did reach a maximum accuracy of above 70%, we can certainly make some more changes and try to improve the accuracy further

0.0.9 7. Improving the model

There are lot of things we can change for improving the model such as: 1. Increasing the number of epochs 2. Adding more convolutional layers 3. Changing the number of filters 4. Changing the size of our filters 5. Changing the pool size of Max Pooling layer

[22]:
```
model_4 = Sequential([
    Input(shape = (224, 224, 3)),
    Conv2D(filters = 10, kernel_size = 2, padding = 'valid', activation =␣
 ↪'relu'),
    MaxPool2D(pool_size = 2),
    Conv2D(filters = 32, kernel_size = 2, padding = 'valid', activation =␣
 ↪'relu'),
    MaxPool2D(pool_size = 2),
    Conv2D(filters = 64, kernel_size = 2, padding = 'valid', activation =␣
 ↪'relu'),
    MaxPool2D(pool_size = 2),
    Flatten(),
    Dense(1, activation = 'sigmoid')
```

```
])

model_4.compile(
    loss = BinaryCrossentropy(),
    optimizer = Adam(learning_rate = 0.0001),
    metrics = ['accuracy'])

history_4 = model_4.fit(
    train_data, # Using non-augmented data
    epochs= 20,
    steps_per_epoch = len(train_data),
    validation_data = val_data,
    validation_steps = len(val_data))
```

```
Epoch 1/20
50/50              12s 157ms/step -
accuracy: 0.5597 - loss: 0.6856 - val_accuracy: 0.5383 - val_loss: 0.6797
Epoch 2/20
50/50              3s 67ms/step -
accuracy: 0.5737 - loss: 0.6689 - val_accuracy: 0.5255 - val_loss: 0.6767
Epoch 3/20
50/50              3s 66ms/step -
accuracy: 0.6563 - loss: 0.6419 - val_accuracy: 0.7117 - val_loss: 0.6236
Epoch 4/20
50/50              4s 72ms/step -
accuracy: 0.7338 - loss: 0.6022 - val_accuracy: 0.7474 - val_loss: 0.5869
Epoch 5/20
50/50              3s 69ms/step -
accuracy: 0.7332 - loss: 0.5722 - val_accuracy: 0.6531 - val_loss: 0.6013
Epoch 6/20
50/50              4s 72ms/step -
accuracy: 0.7490 - loss: 0.5458 - val_accuracy: 0.7296 - val_loss: 0.5451
Epoch 7/20
50/50              3s 67ms/step -
accuracy: 0.7834 - loss: 0.5169 - val_accuracy: 0.7245 - val_loss: 0.5471
Epoch 8/20
50/50              4s 77ms/step -
accuracy: 0.7713 - loss: 0.5061 - val_accuracy: 0.7653 - val_loss: 0.5189
Epoch 9/20
50/50              3s 66ms/step -
accuracy: 0.7922 - loss: 0.4879 - val_accuracy: 0.7730 - val_loss: 0.5020
Epoch 10/20
50/50              3s 67ms/step -
accuracy: 0.7916 - loss: 0.4780 - val_accuracy: 0.7653 - val_loss: 0.4997
Epoch 11/20
50/50              4s 72ms/step -
accuracy: 0.8043 - loss: 0.4605 - val_accuracy: 0.7959 - val_loss: 0.4929
Epoch 12/20
```

```
50/50              3s 62ms/step -
accuracy: 0.8056 - loss: 0.4545 - val_accuracy: 0.7730 - val_loss: 0.4975
Epoch 13/20
50/50              4s 71ms/step -
accuracy: 0.8050 - loss: 0.4490 - val_accuracy: 0.7883 - val_loss: 0.4866
Epoch 14/20
50/50              4s 70ms/step -
accuracy: 0.8183 - loss: 0.4302 - val_accuracy: 0.7857 - val_loss: 0.4920
Epoch 15/20
50/50              4s 71ms/step -
accuracy: 0.8139 - loss: 0.4261 - val_accuracy: 0.7679 - val_loss: 0.4979
Epoch 16/20
50/50              4s 76ms/step -
accuracy: 0.8024 - loss: 0.4383 - val_accuracy: 0.7934 - val_loss: 0.4833
Epoch 17/20
50/50              4s 73ms/step -
accuracy: 0.8266 - loss: 0.4162 - val_accuracy: 0.7934 - val_loss: 0.4829
Epoch 18/20
50/50              4s 76ms/step -
accuracy: 0.8386 - loss: 0.4023 - val_accuracy: 0.7985 - val_loss: 0.4848
Epoch 19/20
50/50              4s 89ms/step -
accuracy: 0.8532 - loss: 0.3923 - val_accuracy: 0.7908 - val_loss: 0.4823
Epoch 20/20
50/50              4s 73ms/step -
accuracy: 0.8488 - loss: 0.3900 - val_accuracy: 0.7883 - val_loss: 0.4864
```

[23]:
```
plot_loss_curves(history_4)
```

So we did managed to improve our accuracy to around 80%, which I think is not that bad considering the amount of images we had.

0.0.10 8. Transfer Learning

[24]:
```
# Data augmentation
data_augmentation = tf.keras.Sequential([
    layers.RandomFlip("horizontal"),
    layers.RandomRotation(0.2),
    layers.RandomZoom(0.2),
])

#EfficientNet langsung
base_model = tf.keras.applications.EfficientNetB0(
    include_top=False,
    weights='imagenet',
    input_shape=(224,224,3)
)

base_model.trainable = False

# Model
inputs = tf.keras.Input(shape=(224,224,3))

x = data_augmentation(inputs)
x = tf.keras.applications.efficientnet.preprocess_input(x)

x = base_model(x, training=False)

x = layers.GlobalAveragePooling2D()(x)
x = layers.BatchNormalization()(x)

x = layers.Dense(128, activation='relu')(x)
x = layers.Dropout(0.5)(x)

outputs = layers.Dense(1, activation='sigmoid')(x)

model = tf.keras.Model(inputs, outputs)

model.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=1e-4),
    loss='binary_crossentropy',
    metrics=['accuracy']
)

history_5 = model.fit(
    train_data,
    validation_data=val_data,
    epochs=20
)
```

```
Epoch 1/20

E0000 00:00:1776065801.355195   16511 meta_optimizer.cc:967] layout failed:
INVALID_ARGUMENT: Size of values 0 does not match size of permutation 4 @ fanin
shape inStatefulPartitionedCall/functional_5_1/efficientnetb0_1/block2b_drop_1/s
tateless_dropout/SelectV2-2-TransposeNHWCToNCHW-LayoutOptimizer

50/50              21s 244ms/step -
accuracy: 0.4898 - loss: 0.7475 - val_accuracy: 0.5000 - val_loss: 0.6937
Epoch 2/20
50/50              8s 168ms/step -
accuracy: 0.5000 - loss: 0.7248 - val_accuracy: 0.5000 - val_loss: 0.6951
Epoch 3/20
50/50              8s 164ms/step -
accuracy: 0.5133 - loss: 0.7249 - val_accuracy: 0.5000 - val_loss: 0.6946
Epoch 4/20
50/50              8s 165ms/step -
accuracy: 0.5133 - loss: 0.7307 - val_accuracy: 0.5000 - val_loss: 0.6934
Epoch 5/20
50/50              8s 163ms/step -
accuracy: 0.4981 - loss: 0.7226 - val_accuracy: 0.5000 - val_loss: 0.6932
Epoch 6/20
50/50              8s 162ms/step -
accuracy: 0.5178 - loss: 0.7190 - val_accuracy: 0.5051 - val_loss: 0.6929
Epoch 7/20
50/50              8s 162ms/step -
accuracy: 0.5133 - loss: 0.7090 - val_accuracy: 0.5051 - val_loss: 0.6928
Epoch 8/20
50/50              8s 163ms/step -
accuracy: 0.5318 - loss: 0.7052 - val_accuracy: 0.4974 - val_loss: 0.6922
Epoch 9/20
50/50              8s 163ms/step -
accuracy: 0.5025 - loss: 0.7219 - val_accuracy: 0.5000 - val_loss: 0.6920
Epoch 10/20
50/50              8s 163ms/step -
accuracy: 0.5445 - loss: 0.6992 - val_accuracy: 0.5281 - val_loss: 0.6912
Epoch 11/20
50/50              8s 163ms/step -
accuracy: 0.5292 - loss: 0.6981 - val_accuracy: 0.5281 - val_loss: 0.6905
Epoch 12/20
50/50              8s 164ms/step -
accuracy: 0.5064 - loss: 0.7139 - val_accuracy: 0.5638 - val_loss: 0.6898
Epoch 13/20
50/50              8s 164ms/step -
accuracy: 0.4930 - loss: 0.7166 - val_accuracy: 0.5714 - val_loss: 0.6890
Epoch 14/20
50/50              8s 164ms/step -
accuracy: 0.5184 - loss: 0.7082 - val_accuracy: 0.5510 - val_loss: 0.6884
Epoch 15/20
```

```
50/50              8s 164ms/step -
accuracy: 0.5133 - loss: 0.7101 - val_accuracy: 0.5510 - val_loss: 0.6872
Epoch 16/20
50/50              8s 164ms/step -
accuracy: 0.5394 - loss: 0.6965 - val_accuracy: 0.5485 - val_loss: 0.6874
Epoch 17/20
50/50              8s 164ms/step -
accuracy: 0.5146 - loss: 0.7038 - val_accuracy: 0.5510 - val_loss: 0.6871
Epoch 18/20
50/50              8s 164ms/step -
accuracy: 0.5426 - loss: 0.6971 - val_accuracy: 0.5408 - val_loss: 0.6873
Epoch 19/20
50/50              8s 164ms/step -
accuracy: 0.5146 - loss: 0.7064 - val_accuracy: 0.5434 - val_loss: 0.6869
Epoch 20/20
50/50              8s 164ms/step -
accuracy: 0.5241 - loss: 0.6983 - val_accuracy: 0.5663 - val_loss: 0.6863
```

[25]:
```
plot_loss_curves(history_5)
```